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Distinguishing placenta accreta from placenta previa via maternal plasma levels of sFlt-1 and PLGF and the sFlt-1/PLGF ratio

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ABSTRACT

Introduction: Our study aimed to distinguish patients with placenta accreta (crete, increta, and percreta) from those with placenta previa using maternal plasma levels of soluble fms-like tyrosine kinase-1 (sFlt-1) and placental growth factor (PLGF) and the sFlt-1/PLGF ratio.
Methods: We obtained maternal plasma from 185 women in late pregnancy and sorted them into three groups: 72 women with normal placental imaging results (control group), 50 women with placenta previa alone (PP group), and 63 women with placenta previa and placenta accreta (PAS group). The concentrations of sFlt-1 and PLGF in the maternal plasma were measured using ELISA kits and the sFlt-1/PLGF ratio was calculated.
Result: The median (min-max) sFlt-1 levels and the sFlt-1/PLGF ratio in the PAS group (12.8 ng/ml, 3.8–34.2 ng/ml) (133, 14-361) were lower than in the PP group (28.7 ng/ml, 13.1–60.3 ng/ml) (621, 156-2013) ($p < 0.0001$ and $P < 0.0001$, respectively). The median (min-max) PLGF levels in the PAS group (108 pg/ml, 38-679 pg/ml) was higher than that in the PP group (43 pg/ml, 12-111 pg/ml) ($p < 0.0001$ and $p < 0.0001$, respectively). The area under the ROC of the sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio were 0.91, 0.90, and 0.99, respectively; the cut-off values were 18.9 ng/ml, 75.9 pg/ml, and 229.5, respectively. The concentration of sFlt-1 and sFlt-1/PLGF ratio were associated with the volume of blood loss (-.288*, -.301*).
Discussion: The concentrations of sFlt-1 and PLGF and ratio of plasma sFlt-1/PLGF may distinguish patients with placenta accreta from those with placenta previa.

1. Introduction

Placenta accreta spectrum disorder (PAS) was recently presented by the International Federation of Gynaecology and Obstetrics [1] and Society of Obstetricians and Gynaecologists of Canada [2]. It refers to the penetration of the trophoblastic tissue through the decidua basalis into the underlying uterine myometrium, the uterine serosa or even beyond, extending to pelvic organs [3]. PAS is associated with adverse maternal and fetal outcomes. Maternal hemorrhage often occurs when the placenta is not properly separated from the uterus after delivery. Severe obstetric and postpartum hemorrhage can cause serious complications and poor outcomes, including disseminated intravascular coagulation, multiorgan failure, hysterectomy, intensive care unit admission, infection, prolonged hospitalization, and even death [4]. Fetal complications are primarily associated with prematurity [5]. Placenta previa is defined as the condition in which the placenta directly overlies the cervix, as opposed to a low-lying placenta, where the placenta is close to the cervical os [6]. Approximately 90% of placenta

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previa cases diagnosed between 15 and 19 weeks of gestation resolve at term [7]. It is necessary to repeat an ultrasonography at ≥32 weeks to reassess the placental location. Transvaginal sonography is the most common method for diagnosing placenta previa, with a sensitivity of 88% and specificity of 99% [6]. Patients with placenta previa alone had better maternal-fetal outcomes than those with placenta accreta. Accurate prepartum diagnosis of placenta accreta, especially in distinguishing placenta accreta from placenta previa can be helpful for thorough preoperative preparation and transfer treatment, decreasing the prevalence of hysterectomy and decreasing intraoperative bleeding and preventing serious intraoperative complications.

As an independent risk factor for placenta accreta, placenta previa can be easily diagnosed [8]; however, patients with placenta previa and placenta accreta (crete, increta, and percreta) are not easily and precisely diagnosed. Currently, the preoperative diagnosis of abnormal placenta accreta mainly depends on prenatal ultrasound and magnetic resonance imaging (MRI). However, neither are not sufficient to predict an abnormally invasive placenta in clinical practice [9,10].

The placenta is an organ that delivers oxygen and nutrients to the fetus [11]. Certain vasoactive proteins, cytokines, proteases, and growth factors contribute to normal placental development and trophoblast invasion, and the fine coordination between placental growth factor (PLGF), vascular endothelial growth factor (VEGF), and soluble fms-like tyrosine kinase-1 (sFlt-1) plays an important role [12–14]. Because trophoblasts invade the decidual tissue and reconstruct the spiral artery, destruction of the invasive function (insufficient depth of invasion) causes placental dysfunction such as fetal growth restriction and preeclampsia [15]. In contrast, an increased depth of trophoblast invasion can lead to abnormally invasive placenta, such as placenta accreta [16]. In preeclampsia, there is insufficient remodeling of the spiral arterioles. Many studies on the pathogenesis of preeclampsia [17–20] have shown that the dynamic balance between levels of sFlt-1 and PLGF is disturbed. Changes in this balance can predict preeclampsia. In PAS, the placental fetal vessels become more elongated and wider with deficient branching [21]. Vessels in the deep myometrium are remodeled, and the maximal depth of the remodeled vessels are increased [22]. This may be associated with abnormal trophoblast invasion. Therefore, we hypothesize that disturbance of the dynamic balance also exists in patients with an abnormally invasive placenta. We measured sFlt-1 and PLGF concentrations in the maternal plasma and calculated the sFlt-1/PLGF ratio in late pregnancy to verify its usefulness as an antenatal predictor of placenta accreta compared with placenta previa alone and normal placentation.

2. Materials and method

2.1. Study design and population

A prospective case-controlled clinical study was performed at the Provincial Hospital. We included 185 pregnant women at 34–41 weeks of late gestation who provided written informed consent during the period between October 2020 and November 2021. Approval was obtained from the local ethics committee.

The study group consisted of 72 women with normal placental imaging results (control group; Group 1), 50 women with placenta previa alone (PP group, Group 2), and 63 women with placenta previa and placenta accreta (crete, increta, and percreta) (PAS group, Group 3). The exclusion criteria included preeclampsia, gestational diabetes, intrauterine growth restriction, pregnant women whose delivery or clinical data were missing, acute and chronic infectious diseases, other surgery and pregnancy complications, and a twin or multiple pregnancy. The prenatal diagnosis of placenta previa or placenta accreta was primarily based on ultrasonography, and MRI was used to diagnose placenta accreta in the posterior wall of the uterus. Placenta accreta (crete, increta, and percreta) was confirmed during surgery.

Peripheral blood (5 mL) from each woman at ≥34 weeks of admission was collected in EDTA. Blood samples were centrifuged at 3000 g for 10 min at 4 °C. The plasma was stored at −80 °C until testing. Plasma sFlt-1 and PLGF leves were tested using an ELISA Kit (BOSTER, Human VEGFR1/sFlt-1 ELISA KIT, EK0543; Bioss, Human PLGF ELISA KIT, bsk11094), and the sFlt-1/PIGF ratio was calculated. Three repeated holes were set in the same sample, and the CV was <5%.

2.2. Statistical analysis

According to whether the continuous variables were normally distributed, we presented continuous variables in this study as the mean ± standard deviation (normal distribution) or median (min-max) (skewed distribution). The Mann–Whitney *U* test and receiver operating curve (ROC) were used to assess significant differences in the sFlt-1 and PLGF levels and sFlt-1/PLGF ratio, and $p < 0.05$ was considered statistically significant. GraphPad Prism 7.0 statistical software was used for analyses. Multinomial logistic regression was used to determine the possible risk factors for placenta accreta, and $p < 0.05$ was considered significant. Pearson's correlation analysis was used to compare the correlation of sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio with the preoperative ultrasound score and intraoperative blood loss. SPSS 22.0 Statistical software was used for all analyses.

3. Results

3.1. Baseline characteristics of the three groups

A total of 185 pregnant women were included in our study: 72 with normal placenta imaging results (control group; Group 1), 50 with placenta previa alone (PP group, Group 2), and 63 women with placenta previa and placenta accreta (crete, increta, and percreta) (PAS group, Group 3). Table 1 lists the demographic characteristics, and clinical histories of the patient. There were no statistically significant differences in age, weight, height, body mass index, or previous uterine cavity operations among the three groups. The gestational age at delivery in the PAS group was significantly lower than that in the control and PP groups ($p = 0.000$ and $p = 0.005$) (Table 1). The gestational age at delivery was lower in the PP group than in the control group ($p = 0.010$) (Table 1). Intraoperative bleeding was significantly higher in the PAS group than in the control and PP groups ($p = 0.000$ and $p = 0.000$) (Table 1). Intraoperative bleeding was higher in the PP group than in the control group ($p = 0.000$) (Table 1). The number of previous caesarean sections in the PAS and PP groups was significantly higher than that in the control group ($p = 0.003$ and $p = 0.008$, respectively) (Table 1). There were no statistically significant differences in previous caesarean sections between the PP and PAS groups ($p = 0.212 > 0.05$) (Table 1). Neonatal weight in the PAS and PP groups was significantly lower than that in the control group ($p = 0.007$ and $p = 0.040$, respectively) (Table 1). There were no statistically significant differences in neonatal weight between the PP and PAS groups ($p = 0.239 > 0.05$) (Table 1).

3.2. Maternal plasma levels of sFlt-1 and PLGF and the sFlt-1/PLGF ratio

Comparing plasma sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio, we found that PLGF levels had a significantly higher value in the PAS group than that in control and PP groups(Table 2, Fig. 1) and was not associated with intraoperative blood loss (Fig. 2B). Additionally, sFlt-1 levels and sFlt-1/PLGF ratio values in PAS group were significantly lower than that in control and PP groups (Table 2, Fig. 1) and were associated with intraoperative blood loss (Fig. 2A and C). There were also differences between the control and PP groups in PLGF levels and the sFlt-1/PLGF ratio (Table 2, Fig. 1). We found no differences among the three plasma markers and preoperative ultrasound scores (Fig. 2).

The results of the ROC analysis determining the cut-offs with the highest sensitivity and specificity for the three groups are presented in Fig. 1 and Table 3. The areas under the ROC for the sFlt-1 levels, PLGF

Table 1
Demographic analysis and clinical history among the three groups.

	control[1] (n = 72)	PP[2] (n = 50)	PAS[3] (n = 63)	P value
Age (years)	31 ± 4	34 ± 5	34 ± 5	P(1-2) = 0.487 P(1-3) = 0.960 P(2-3) = 0.292
Height (cm)	163 ± 5	162 ± 4	160 ± 4	P(1-2) = 0.170 P(1-3) = 0.096 P(2-3) = 0.377
Weight (kg)	74 ± 10	73 ± 10	71 ± 11	P(1-2) = 0.184 P(1-3) = 0.131 P(2-3) = 0.508
BMI (kg/m^2)	27.9 ± 3.6	27.8 ± 4.0	27.7 ± 3.9	P(1-2) = 0.188 P(1-3) = 0.126 P(2-3) = 0.474
Gestational age (weeks)	39.2 ± 1.3	37.1 ± 1.7	35.7 ± 1.1	**P**(1-2) = 0.010* **P**(1-3) = 0.000* **P**(2-3) = 0.005*
Uterine cavity operations	0(0–4)	1(0–3)	1(0–4)	P(1-2) = 0.987 P(1-3) = 0.295 P(2-3) = 0.083
Previous caesarean sections	23(31.95%)	26(52%)	58(92.10%)	**P**(1-2) = 0.008* **P**(1-3) = 0.003* P(2-3) = 0.212
Intraoperative bleeding	300 (120–500)	500 (200–1500)	1000 (300–8000)	**P**(1-2) = 0.000* **P**(1-3) = 0.000* **P**(2-3) = 0.000*
Neonatal weight (g)	3390 ± 364	2955 ± 506	2790 ± 431	**P**(1-2) = 0.040* **P**(1-3) = 0.007* P(2-3) = 0.239

Group 1: normal placenta imaging results (control group).
Group 2: placenta previa alone (PP group).
Group 3: placenta previa and placenta accreta (PAS group).
BMI: body mass index.
Normal distribution of data is presented as the mean ± standard deviation.
skewed distribution of data is presented as the median (min-max).
Categorical data are expressed as numbers and percentages (n, %).
The results were analyzed by multinomial logistic regression.
P value: between groups; *p < 0.05 significantly different.
Statistically significant p values are marked in bold text.

levels, and sFlt-1/PLGF ratio for discriminating the PAS group from the control group were 0.92, 0.93, and 0.98, respectively. The areas under the ROC for the sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio for discriminating between the PAS and PP group were 0.91, 0.90, and 0.99, respectively. The areas under the ROC for the sFlt-1 levels, PLGF levels and sFlt-1/PLGF ratio for discriminating between the PP and control groups were 0.60, 0.65, and 0.66, respectively (Fig. 1). ROC comparison analysis showed significantly greater specificity and sensitivity of the sFlt-1/PLGF ratio than of the PLGF and sFlt-1 levels for PAS diagnosis. The cut-off values of the sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio for the three groups are presented in Table 3.

Table 2
Circulating levels of plasma sFlt-1and PLGF and the sFlt-1/PLGF ratio in the three groups.

Parameter	control[1] (n = 72)	PP[2] (n = 50)	PAS[3] (n = 63)	P value
sFlt-1 (ng/ml)	32.7 (8.9–66.6)	28.7 (13.1–60.3)	12.8 (3.8–34.2)	P(1-2) **P**(1-3) **** **P**(2-3) ****
PLGF (pg/ml)	33 (6–155)	43 (12–111)	108 (38–679)	**P**(1-2)** **P**(1-3) **** **P**(2-3) ****
sFlt-1/PLGF ratio	953 (170–7239)	621 (156–2013)	133 (14–361)	**P**(1-2)** **P**(1-3) **** **P**(2-3) ****

Data expressed as median (min-max).
Results analyzed by the Mann-Whitney *U* test.
P value: between groups; *p < 0.05 significantly different.
Statistically significant p values are marked in bold text.

4. Discussion

PAS is often associated with massive postpartum hemorrhage and commonly leads to emergency hysterectomy. Placenta previa alone does not cause severe adverse effects. It is particularly important to accurately predict PAS preoperatively, especially to distinguish PAS from placenta previa. A previous study showed that placental bulge was an independent predictor of severe PAS on ultrasound. In diagnosing severe PAS, the placental bulge sign was achieved on ultrasound with an accuracy of 85.5%, sensitivity of 91.7%, and specificity of 76.9%, with an accuracy of 90.3%, sensitivity of 94.4%, and specificity of 84.6% on MRI [23]. One meta-analysis showed that the sensitivity and specificity of ultrasound for clearly diagnosing PAS are 90.72% and 96.94%, respectively. Color Doppler has the best predictive accuracy, with a sensitivity of 90.74% and specificity of 87.68% [24]. Another meta-analysis showed that the sensitivity and specificity of MRI for clearly diagnosing PAS are 94.4% and 84%, respectively [25]. There was no clear difference in the accuracy between ultrasonography and MRI. However, the results may have been affected by selection bias (high risk factors were previously assessed by an ultrasonographer) and operator experience. Simple and reliable biomarkers can help diagnose PAS using ultrasonography before surgery.

Although the pathophysiological and epidemiological risk factors for PAS are widely known [26], the underlying mechanisms leading to an abnormally invasive placenta are poorly understood [27]. Some studies [22,28,29] have shown that an abnormally invasive placenta arises primarily from factors such as reduced, absent, or abnormal decidua, or defective decidualization, leading to an increased depth of penetration by (normal) extravillous trophoblasts. Trophoblast-specific elements, such as growth-, angiogenesis-, and invasion-related factors, might play an important role in the process of increased trophoblast invasiveness [16]. There were two distinct types of vessel growth. One is vasculogenesis, which is the formation of new blood vessels from hemangiogenic stem cells derived from mesenchymal cells that differentiate into hemangioblastic stem cells. Vascular endothelial growth factor (VEGF) and vascular endothelial growth factor receptors (VEGFR) play an important role in this process. Angiogenesis is the formation of new

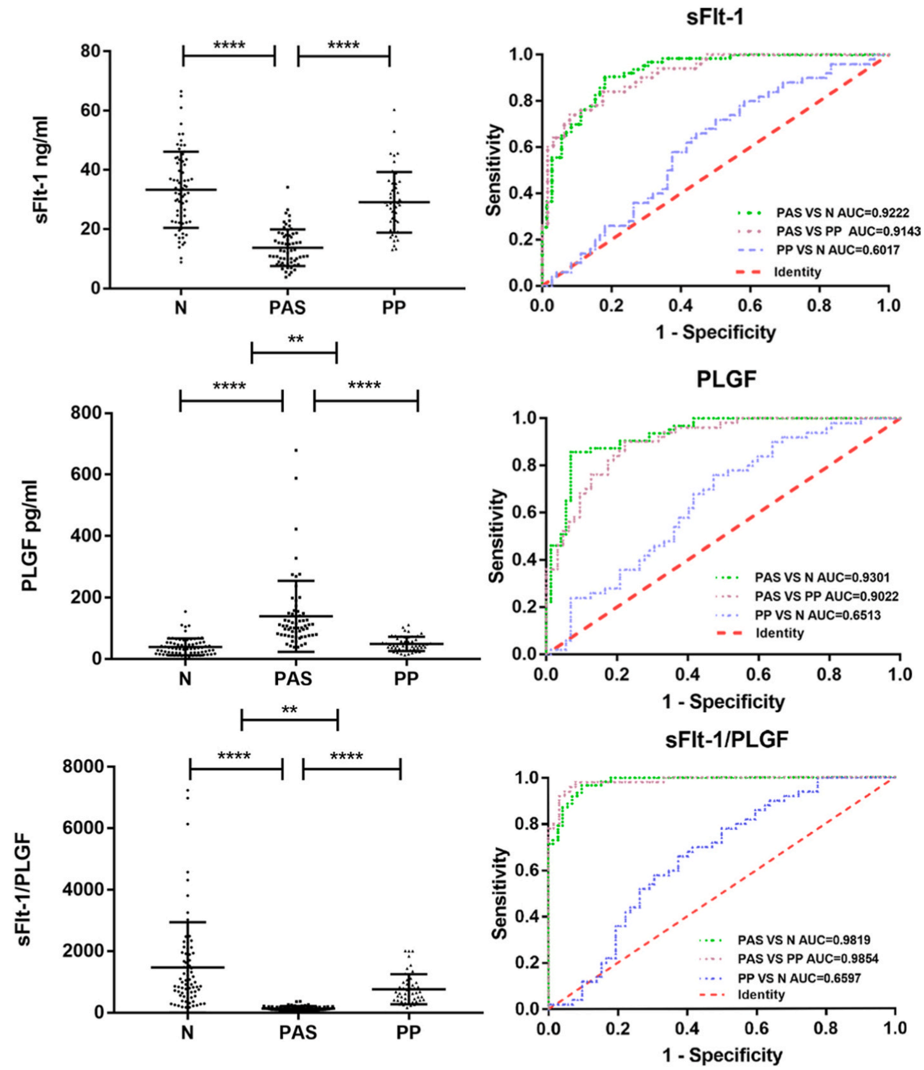


Fig. 1. Circulating levels of the plasma sFlt-1and PLGF and the sFlt-1/PLGF ratio in the three groups. The ROC showing the predictive values of sFlt-1levels, PLGF levels, and sFlt-1/PLGF ratio in the three groups.

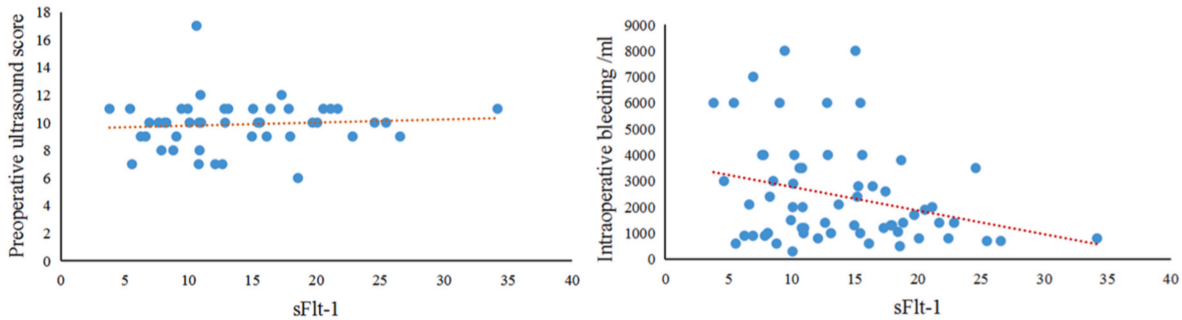
branches from pre-existing vessels, and PLGF is involved in this step. The expression of VEGF and angiogenic growth factors mediates placental development via a series of endocrine and paracrine signals, which begin with the infiltration of cytotrophoblasts in the endometrium [30,31]. VEGF and PLGF are members of the VEGF family and have synergistic effects on angiogenesis. sFlt-1 combines irreversibly with these factors to prevent PLGF and VEGF function [32]. VEGF and PLGF have potent angiogenic characteristics, while sFlt-1 has antiangiogenic potential [30,33].

The scar tissue of the uterus caused by a caesarean section is relatively anoxic, which is mainly due to a local increase in fibrous tissue, absence of re-epithelialization, decreased angiogenesis, and impaired blood circulation around the scar [34–36]. During cytotrophoblast invasion, trophoblasts differentiate and change their behavior once they reach the spiral arterioles, and the spiral arterioles reorganize to increase oxygen tension and delivery [37]. The relative hypoxia of the caesarean scar tissue can preferentially recruit the blastocyst to implant in areas that result in an increased risk of placenta accreta [38]. Due to scar dehiscence, trophoblasts have better access to large outer myometrial vessels and invade them [22].

VEGF, PLGF, and their receptors play integral roles in promoting vessel formation and establishment. sFlt-1 is an alternatively spliced and truncated version of the VEGF receptor 1 (VEGF-R1) and contains an extracellular ligand-binding domain but lacks the transmembrane and cytoplasmic portions of VEGF-R1. sFlt-1 can bind to both VEGF and PLGF to inhibit its angiogenic function [39,40]. During normal pregnancy, VEGF is an angiogenic inducer that promotes angiogenesis under hypoxic induction [41]. PLGF is a VEGF homologue with approximately 50% homology to VEGF. They have two cognate receptors, VEGF-R1 and VEGF receptor 2, known as fms-like tyrosine kinase-1 (Flt-1) and kinase domain region (KDR). KDR is responsible for the action of VEGF on endothelial cells. PLGF does not bind to the KDR receptor, but binds to the Flt-1 receptor with high affinity. PLGF acts by displacing VEGF from the Flt-1 receptor, allowing it to bind to the active KDR receptor and enhance the pro-angiogenic effect of VEGF [42,43].

In our study, we found that plasma sFlt-1 levels and the sFlt-1/PLGF ratio of the PAS group were significantly lower than those of the control and PP groups, and the level of PLGF was the reverse. In addition, we found that sFlt-1 concentration and sFlt-1/PLGF ratio were associated to the volume of blood loss (Pearson correlation analysis -.288*, -.301*). Blood loss may be associated with excessive trophoblast invasion of the great vessels of the deep myometrium in PAS patients. Moreover, in the ROC analyses, the area under the sFlt-1/PLGF ratio curve was close to 1, indicating a better prediction value. Although the plasma PLGF levels and sFlt-1/PLGF ratios differed between PP and control groups, the area under the ROC curve was less than 0.7. We believe that PLGF levels and the sFlt-1/PLGF ratio cannot accurately discriminate between PP and control groups. In Uyankoglu's study, serum sFlt-1, VEGF, and PLGF

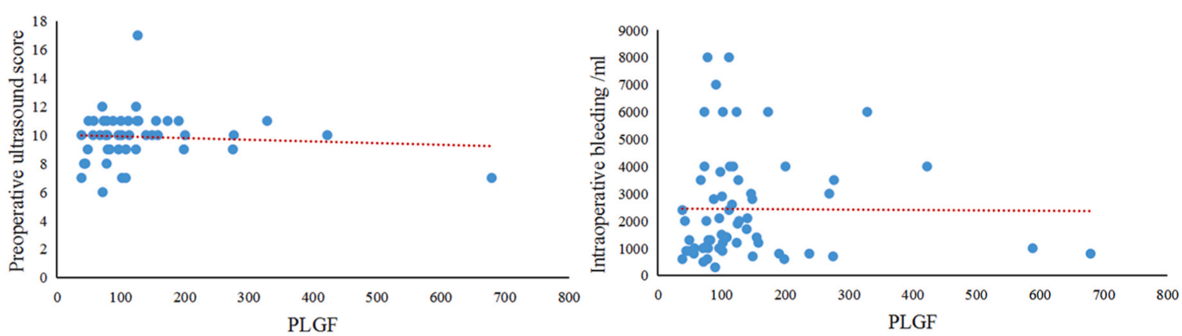
A



	sFlt-1	Preoperative ultrasonographic score	Intraoperative bleeding/ml
sFlt-1	1		
Preoperative ultrasonographic score	0.083	1	
Intraoperative bleeding/ml	-.288*	-.339*	1

*at level 0.05(two-tailed),the correlation was significant

B



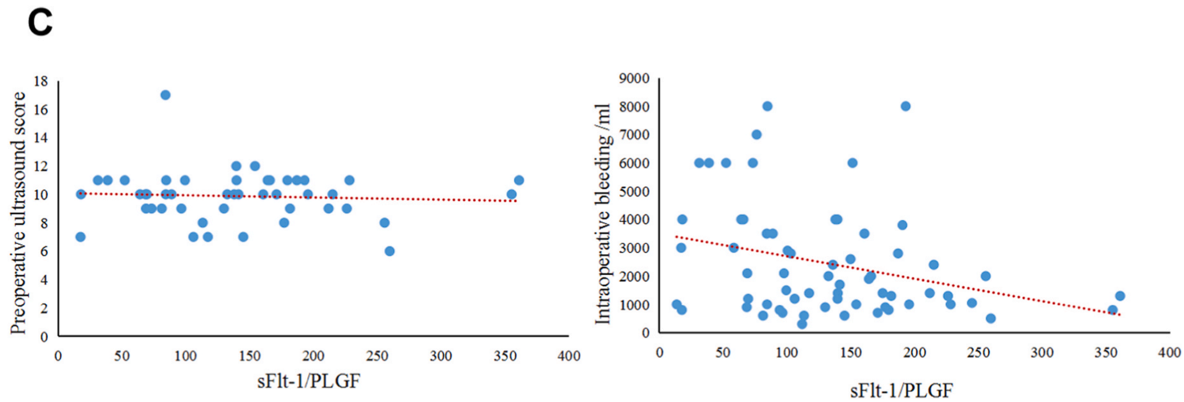
	PLGF	Preoperative ultrasonographic score	Intraoperative bleeding/ml
PLGF	1		
Preoperative ultrasonographic score	-0.075	1	
Intraoperative bleeding/ml	-0.008	-.339*	1

*at level 0.05(two-tailed),the correlation was significant

Fig. 2. The correlations between sFlt-1 levels, PLGF levels, and sFlt-1/PLGF ratio and preoperative ultrasound score and intraoperative bleeding. Results from Pearson's correlation analysis. P value: between groups; *p < 0.05 significantly different.

levels in the placenta previa accreta group were lower than those in the control group [44]. In another study by Biberoglu, serum sFlt-1, VEGF, and PLGF levels were not different between placenta accreta and control groups [45]. A study in China found that PLGF multiples of the median in the placenta previa and accreta group was higher than that in the control group [46]. These different results may be due to the different sample sources, testing methods, and gestational weeks.

In conclusion, the concentrations of sFlt-1 and PLGF and the sFlt-1/PLGF ratio in maternal plasma may be applicable as biological markers for predicting placenta accreta, and as a more effective means to assist



	sFlt-1/PLGF	Preoperative ultrasonographic score	Intraoperative bleeding/ml
sFlt-1/PLGF	1		
Preoperative ultrasonographic score	0.067	1	
Intraoperative bleeding/ml	-0.301*	-0.339*	1

*at level 0.05(two-tailed),the correlation was significant

Fig. 2. (*continued*).

Table 3
Sensitivity, specificity, 95% CI, and the cut-off values within the same group.

		Sensitivity	Specificity	95% CI	Cut-off
sFlt-1	PAS vs. N	0.90	0.82	0.88 to 0.97	21.8 ng/ml
	PAS vs. PP	0.84	0.82	0.86 to 0.96	18.9 ng/ml
	PP vs. N	0.72	0.50	0.50 to 0.70	33.1 ng/ml
PLGF	PAS vs. N	0.86	0.93	0.89 to 0.97	69.8 pg/ml
	PAS vs. PP	0.90	0.78	0.85 to 0.96	75.9 pg/ml
	PP vs. N	0.76	0.53	0.55 to 0.75	33.6 pg/ml
sFlt-1/PLGF	PAS vs. N	0.97	0.90	0.97 to 1.00	267.9
	PAS vs. PP	0.98	0.92	0.97 to 1.00	229.5
	PP vs. N	0.66	0.63	0.56 to 0.76	795.0

ultrasound in the preoperative diagnosis of placenta accreta. Serum concentrations of sFlt-1 and PLGF and the ratio of plasma sFlt-1/PLGF can distinguish patients with placenta accreta (crete, increta, and percreta) from patients with placenta previa. Our study was limited by the small sample size and gestational age at delivery; therefore we could not determine the value of these biological markers in predicting the occurrence of placenta accreta in early pregnancy. The association of sFlt-1 levels, PLGF levels and the sFlt-1/PLGF ratio with different grades of placenta accreta and placenta accreta complicated by other diseases associated with pregnancy was not investigated. Further studies are required to confirm our results.

Author's contributions

F.Y. Z., Y. L., and L. L. conceived and designed the study. F. Y. Z. performed experiments and drafted the manuscript. M. Q. G. collected clinical specimens, P. Z. C. and S. T. W. performed statistical analysis. L. L., Y. L. and Q. Z. revised the manuscript. * Two corresponding authors contributed equally to this work. All the authors have read and approved the final manuscript.

Ethics approval and consent to participate

The study was approved by the Human Ethics Committee of the Provincial Hospital Affiliated to Shandong First Medical University, Jinan, 250021 China.

Data statement

All data can be obtained from the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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